

String name = "Aditya"

int age = 28;

int rollno = 101;

String college = "IIT Guwahati";

String name = "Rohit"

int age = 28;

int rollno = 102;

String college = "IIT Guwahati";

void print(String name, int age, int rollno, String college)

{
 Sop(---);
}

problems

(1) Need a lot of independent variable to represent one student

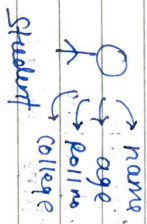
(2) If I pass the student to a function I need to pass all 4 variables.

(3) I need to create new set of variables for new student.

(4) No authority over data

Object oriented programming

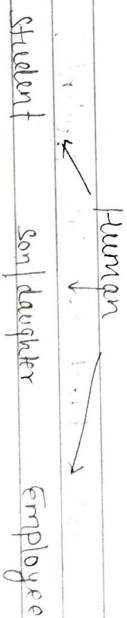
⇒ Real world → mimic



name	age
rollno	college

Real world

programming world



Real world

programming

student

- name
- age
- rollno
- college

```

class student {
    string name
    int age
    int rollno
    string college
}
    
```

⇓
Blueprint of an object.

Q How to create object of class student?

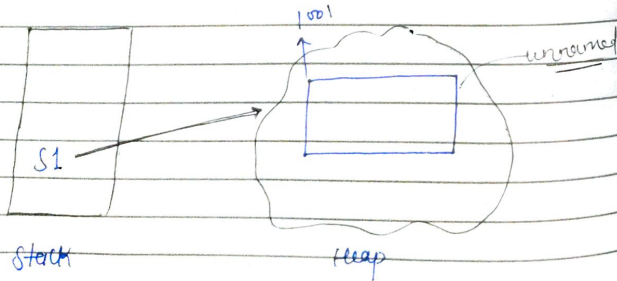
Ans

Student s1; // declaration

s1 = new student(); //

or

student s1 = new student();



→ S1 → store starting address of object.
↳ reference variable

→ "S1 is not object"

→ new keyword work at run time.
↳ dynamic memory allocation.

→ ① Compile time (static)

② Runtime (Dynamic)

Now,

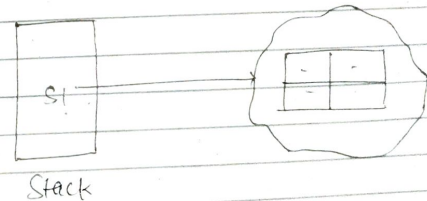
Student s1 = new student();

s1.name = "Aditya";

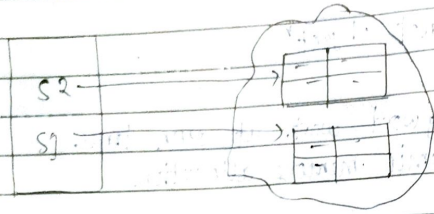
s1.rollno = 101;

s1.age = 28;

s1.college = "IIT Guwahati";



Student s2 = new Student();



s2.name = "Marish";

s2.age = 20;

s2.rollno = 103;

s2.college = "XYZ";

print as;

System.out.println(s2.name);

Result

Marish.

Naming Convention (Good practice);

① Variables (primitive / reference variable).

↳ Camel case.

Eg.

firstName, lastName
firstNameOfStudent.

② classes — starting with capital letter.

Eg.

Student, Human, Character.

③ Functions → camel case.

⇒ Animal

↳ characteristics (type, no of legs - -)

↳ Behaviour (make sound, eat sun - -)

class Animal {

String type;

int noOfLegs;

void makeSound() {

=

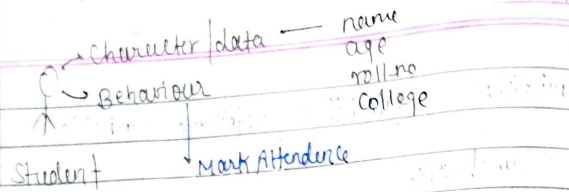
}

void eat() {

=

}

}



```

class Student () {
    String name;
    int age;
    int rollNo;
    String College;
}
    
```

Character / data

```

void markAttendance () {
    // ...
}
    
```

Behaviour

```

Student s1 = new Student ();
s1.name = "John";
s1.markAttendance ();
    
```

```

void print (Student s) {
    System.out.println(s.name, s.age);
}
    
```

→ we can add print function inside class.

```

class Student {
    // ...
    void print () {
        // (name, age, college, roll no);
    }
}
    
```

Real Object → Digital Object

Animal →	✓
Student →	✓
Bank Account →	✓
Location →	✓

```

class BankAccount {
    acctNo;
    IFSC;
    branch;
}
    
```